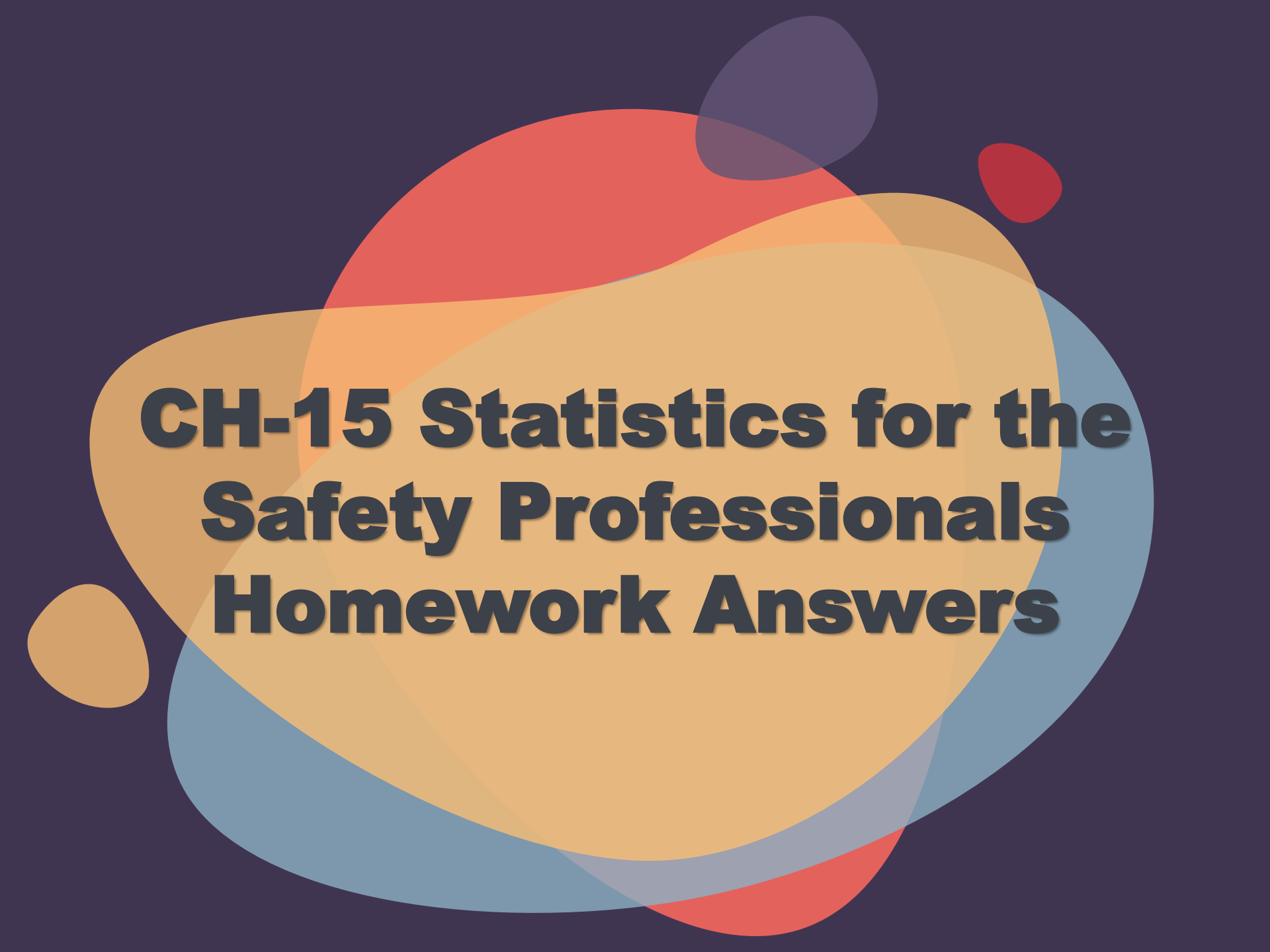




CH-15 Statistics for the Safety Professionals Homework Answers



01

What is the mean of a group of safety practitioners who sit for the ASP exam and receive the following scores: 76, 88, 94, 55, 75, 85, 88, 64, 82, 73, 72, 70?

A. 67

B. 77

C. 76

D. 92

$$\mu = \bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$76 + 88 + 94 + 55 + 75 + 85 + 88 + 64 + 82 + 73 + 72 + 70 = 922$$

$$\text{Mean} = 922 \div 12 = 76.8 \approx 77$$

02

Calculate the standard deviation using the following sample:
7, 12, 18, 5, 6, 6, 14, 11 and 19.

You will need the following formula:

$$\sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$$

A. 5.1

B. 4.3

C. **5.3**

D. 0.056

X	$\bar{X}$	$X - \bar{X}$	$(X - \bar{X})^2$
7	10.98	-3.89	15.1321
12	10.98	1.11	1.2321
18	10.98	7.11	50.5521
5	10.98	-5.89	34.6921
6	10.98	-4.89	23.9121
6	10.98	-4.89	23.9121
14	10.98	3.11	9.6721
11	10.98	0.11	0.0121
19	10.98	8.11	65.7721
ΣX			$\Sigma(X - \bar{X})^2$
98			= 224.8889

$$= \sqrt{(224.89/8)}$$

$$= 5.3$$

03

The following table shows estimates of the error rates within the project development phases of a system. What is the best estimate of the mean error rate?

Error Rates Within Project Development Phases	
Phase	Errors Per 100 Object Instructions
Validation and Acceptance	0.768
Integration Test	0.504
Operational Demonstration	0.371
Pre-operations Exercise	0.337

A. 1 error per 50 instructions

B. 1 error per 200 instructions

C. 1 error per 1000 instructions

D. 1 error per 2000 instructions

$$0.768 + 0.504 + 0.371 + 0.337 = 1.98$$

$$1.98 \text{ errors/400 instructions} \approx 1 \text{ error / 200 instructions}$$

04

A company has maintained incident records for several years. The mean annual number of incidents has been 28, with a standards deviation of 12. What is the probability of more than 16 incidents next year?

$$Z = \frac{x - \mu}{\sigma}$$

Score (x) Mean (μ) SD (σ)

$$Z = \frac{X - \bar{X}}{\sigma} = \frac{16 - 28}{12} = \frac{-12}{12} = -1$$

A. 95%

B. 68%

C. 16%

D. 84%

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611

P-value from Z-Table:

$$P(x < 16) = 0.1587$$

$$P(x > 16) = 1 - P(x < 16) = 0.8413 = \mathbf{84.13\%}$$



05

The term used to describe the extent to which values spread on either side of a mean, median or mode is called:

A. Limit

B. Range

C. Deviation

D. Dispersion

06

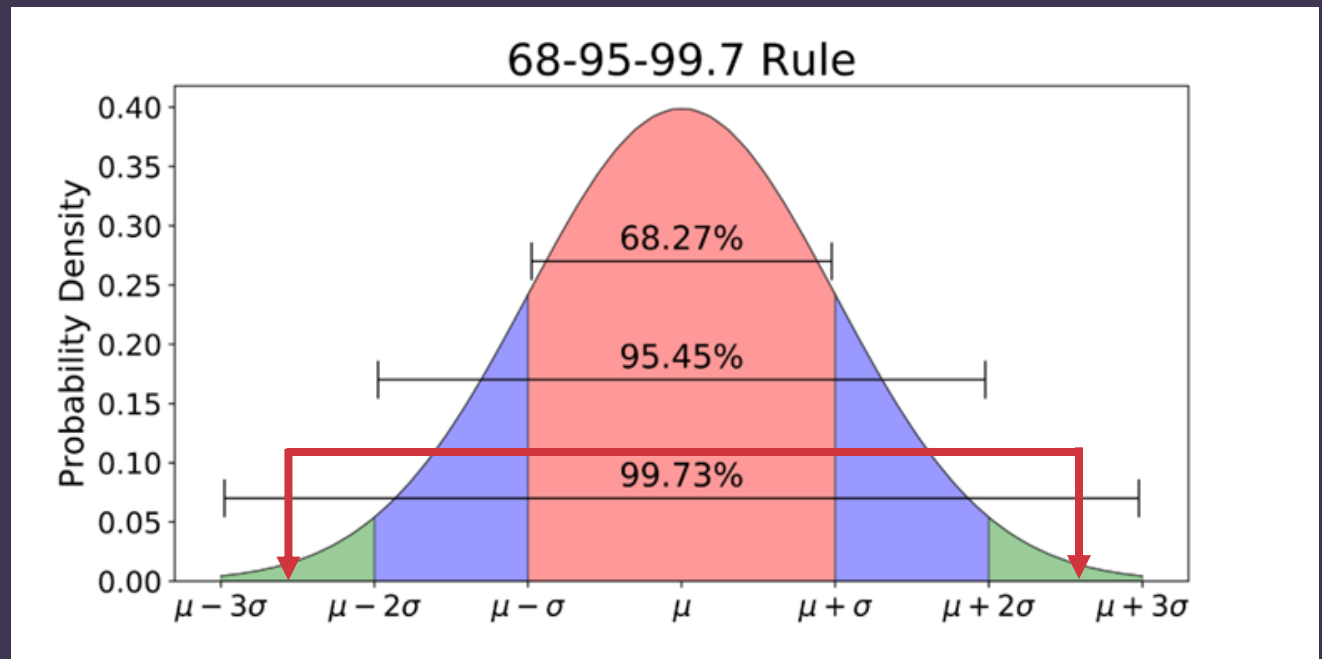
In a normal distribution 99% of the observations will fall within approximately ____ standard deviation(s) of the mean.

A. 1.96

B. 2.58

C. 3.00

D. 4.40





07

A number that expresses the distance from a particular value (X) above or below the mean (μ) in units of standard deviation (S), is called _____. The result of this calculation is used in statistical tests involving normal distributions.

A. Variable

B. Z-Score

C. Variance

D. Quartile

08

The correlation formula is used to evaluate the relationship between two variables of the same interval. Select the best correlation.

A. 0.05

B. 0.50

C. 0.92

D. 1.006

$$1- \geq r \leq +1$$

09

In a semi-conductor operation, a random sampling of 100 wafers showed that 3 were faulty. In a sampling of 2500 wafers, how many may be expected to be faulty?

A. 30

B. 75

C. 100

D. 150

There is a probability of $3/100 = 3\%$ that a wafer may be faulty.

$$(3\%)(2500) = 75$$

$$\begin{array}{cc} 3 & 100 \\ x & 2500 \end{array}$$

$$x = \frac{3 \times 2500}{100} = 75$$



10

In a series reliability model all components must work, the system reliability is a function of all the component reliabilities. The appropriate equation is:

A. $R_{(s)} = 1 - R_{(a)} + 1 - R_{(b)} + \text{etc}$

B. $R_{(s)} = R_{(a)} \times R_{(b)} \times \text{etc}$

C. $R_{(s)} = R_{(a)} - R_{(b)} + 1$

D. $R_{(s)} = R_{(a)} + (1 - R_{(b)}) + (1 - R_{(c)}) + \text{etc}$



11

A switch is operated twice per day. If the switch fails after ten years of operation, what is its failure rate?

Total # of operations = (10 years) (365 days/year) (2 operations/day)
= 7300 operations.

Therefore 1 failure/7300 operations = 0.00014 = 1.4×10^{-4}

A. 2.7×10^{-6}

B. 1.8×10^{-5}

C. 1.4×10^{-4}

D. 1.9×10^{-7}



12

A switch has 5 components. Each part has a failure rate of 0.3. Failure of any part causes the switch to fail. What is the reliability of the switch system?

A. 1.50

B. 0.168

C. 0.150

D. 0.00243

Reliability is defined as (1 - failure rate).

The reliability of this system is the product of the reliabilities of the components.

$$R = 1 - 0.3 = 0.7$$

$$R_{(S)} = R_1 \times R_2 \times R_3 \times R_4 \times R_5$$

$$R = (1 - 0.3)^5 = (0.7)^5 = \mathbf{0.168}$$